

TAMIL NADU STATE BOARD - STANDARD 12 MATHEMATICS VOLUME 1

CHAPTER 6: VECTOR ALGEBRA

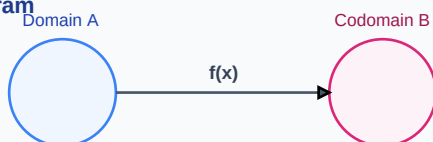
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Mathematics Volume 1
Chapter Name:	Vector Algebra

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Scalar Triple Product	The dot product of a vector with the cross product of two other vectors, denoted by $[a, b, c]$, representing parallelepiped volume.
Vector Triple Product	The cross product of a vector with the cross product of two other vectors, yielding a vector perpendicular to both vectors.
Skew Lines	Lines in three-dimensional space that are not parallel and do not intersect, lying on different planes.

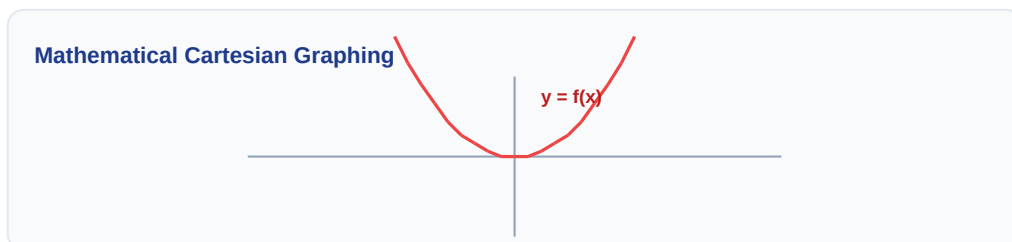
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Scalar Triple Product: $[a, b, c] = a \cdot (b \times c)$ | $[a, b, c] = 0$ means coplanar
- Vector Triple Product Expansion: $a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$
- Shortest distance (skew lines): $d = |(a_2 - a_1) \cdot (u \times v)| / |u \times v|$
- Plane Equation: $r \cdot n = d$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find volume of a parallelepiped whose coterminous edges are $i + j$, $2i - k$, and $j + k$.

Step-by-step Solution:

$$a = i+j, b = 2i-k, c = j+k.$$

$$[a, b, c] = \det\left(\begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}\right) = 1(0 - (-1)) - 1(2 - 0) + 0 = 1 - 2 = -1.$$

$$\text{Volume} = |-1| = 1 \text{ cubic unit.}$$

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Find the vector and cartesian equations of the plane passing through points (2, 2, 1) and (9, 3, 6) and perpendicular to plane $2x + 6y + 6z = 9$.

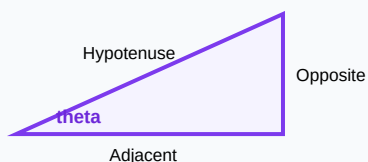
Analytical Solution Strategy:

Normal vector n_1 of target plane must be perpendicular to plane vector (7, 1, 5) and normal of given plane (2, 6, 6).

$n_1 = (7, 1, 5) \times (2, 6, 6) = (-24, -32, 40)$ proportional to (3, 4, -5).

Plane passing through (2, 2, 1): $3(x-2) + 4(y-2) - 5(z-1) = 0 \Rightarrow 3x + 4y - 5z = 9$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find shortest distance between parallel lines $r = a_1 + t \cdot u$ and $r = a_2 + s \cdot u$.

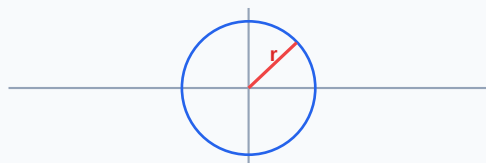
Q2 (2-Mark): Evaluate coplanarity of vectors: $2i - j + k$, $i + 2j - 3k$, $3i + j - 2k$.

Q3 (2-Mark): Write down vector equation of a line passing through $(1, 2, 3)$ in direction of $2i - j + 2k$.

Q4 (5-Mark): Define vector triple product.

Q5 (5-Mark): Find angle between the planes $r \cdot (i + j + k) = 5$ and $r \cdot (2i - j + k) = 3$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Aviation Traffic

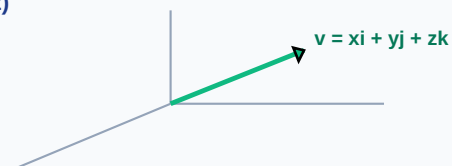
Collision avoidance algorithms calculate the shortest distance between plane path lines using skew line formulas.

Application Case: 3D Mesh Rendering

Illumination angles are resolved by calculating plane normal vectors on face meshes.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

